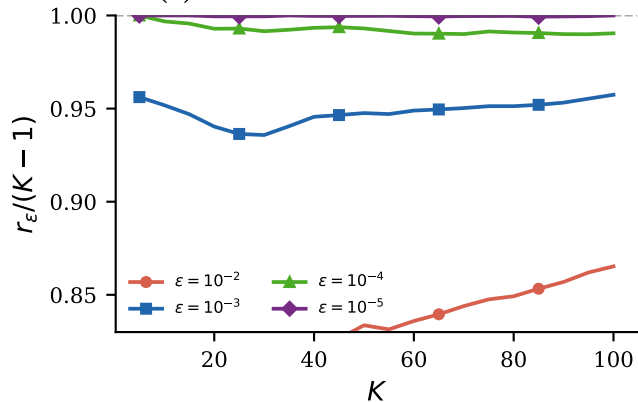
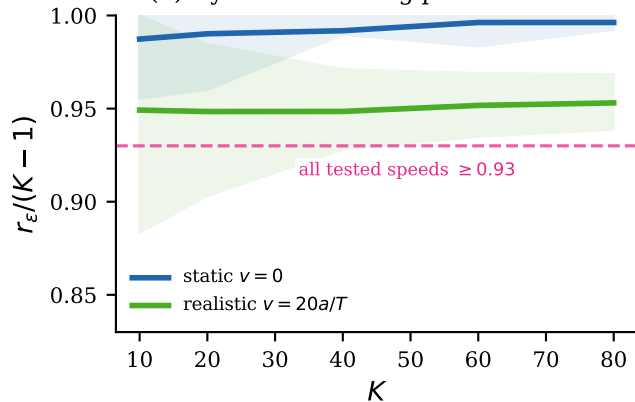
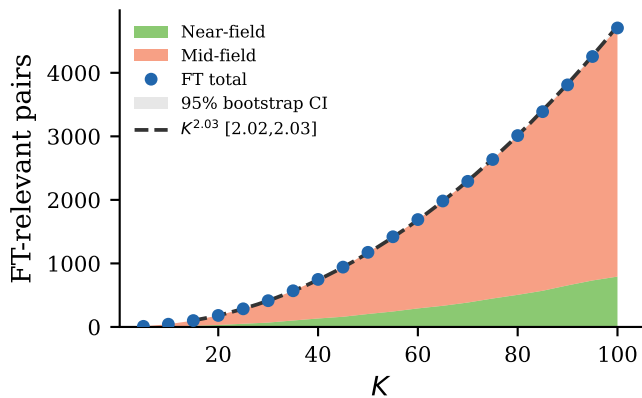


(a) Near-full rank across thresholds



(b) Dynamic shuttling preserves rank

(c) Pair count scales as $O(K^2)$
($X_{\max} = 20a$, 95% of pairs)

(d) Pairwise calibration drift wall

